

Quantitative Market Risk



- Market Risk Overview
- VaR Models
 - Simple VaR Model for S&P500
 - Types of VaR Models
 - Using the right historical data –in the right way
 - Portfolio Recomputation
 - Capital Allocation
 - General Modelling Challenges
- Fundamental Review of the Trading Book (FRTB)
 - Overview
 - Standardized Approach
 - Delta and Vega Capital
 - Curvature Capital
 - Default Charge
 - RRAO
 - Internal Model Approach
 - Stress Scaling and IMCC Aggregation
 - Capital Calculation
 - Backtesting
 - P&L Attribution test

- Market Risk: Potential loss in the value of a risky asset or portfolio over a defined period of given confidence interval

- Key Consumers of the output:
 - Internal: risk managers, senior management, finance
 - External: regulators, supervisors, investors, outside analysts

- Key Elements:
 - Confidence level, e.g. 95% or 99% (for VaR and SVaR) and 99.9% for IRC and CRM
 - Shock sizes, e.g. 1-day or 10-day
 - VaR calculation window, e.g. 1-year
 - Simulation : Historical Simulation (General Market Risk) and Monte Carlo (Specific / Default Risk)

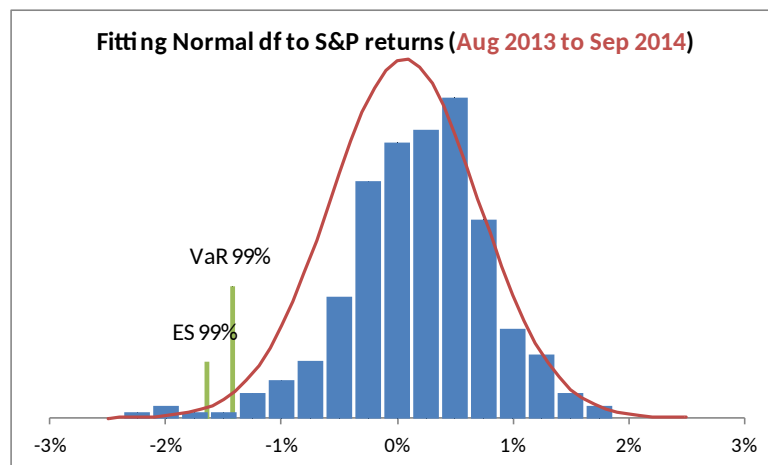
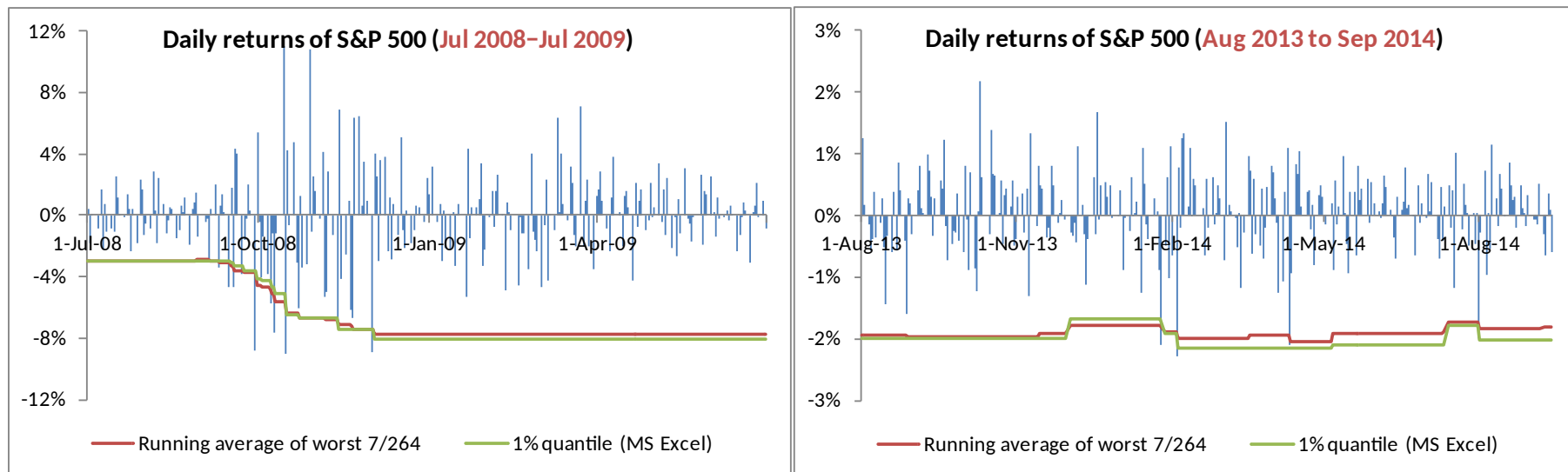
VaR Models

A simple VaR model for the S&P 500

Imagine a unit investment in the S&P 500. Our objective is to characterize the downside risk over a one-day horizon.

Few Conventions for Historical VaR:

- prediction is based on daily returns for a running look-back period equal to 264 trading days (≈ 1 calendar year)
- use the average of the worst-7 outcomes as a proxy* for the 1% quantile (*this corresponds to the assumption that $ES_{97.35\%} = VaR_{99\%}$, which holds true for the Normal distribution)



The empirical return distribution shows evidence of asymmetry and tails that are fatter than those of the normal distribution.

Remark: VaR and ES in the graph are those of the depicted Normal distribution; not necessarily those of the data.

Types of VaR models

A distributional assumption for the risk factors is needed to simulate from, or characterize, the loss distribution

Common approaches include:

- *Historical simulation*

Apply past returns to today's portfolio

- Advantages: Non-parametric; no calibration required; results are linked to historical events
- Disadvantages: Limited look-back window (data rolls off); historical samples are inherently small; unsuitable in the presence of 'sticky' market data

- *Variance-Covariance method*

If a tractable joint distribution is chosen for the risk factors, the loss distribution may become analytically known

- Advantages: Useful in simple contexts or for prototyping; no simulation noise
- Disadvantages: Not applicable for all risk factor distributions or loss profiles

- *Monte Carlo simulation*

Generate samples from any risk factor distribution

- Flexibility in the choice of distributions and event types
- Introduces the ability to use both Conditional and Unconditional risk factor distributions

Using the right historical data –in the right way

A common theme in using historical data to calibrate a simulation model is to remove the dependency on the then-pertaining market environment in past returns. This manifests itself in two ways: *which data* to use and *how* to use it.

■ Choice of relevant historical data

- Example: How capture implied volatility risk for an equity call option with strike K and time to maturity T ?
 - Use historical data of σ for strike= K and time-to-maturity= T (*constant strike*)
 - Use historical data of σ for moneyness= (S/K) and time-to-maturity= T (*constant moneyness*)
- At a philosophical level, this problem is ubiquitous when dealing with historical data
 - The volatility of, and correlation between, risk factors may depend on state variables, most of which may not be known to us:
 - Calibrating the model to a long history helps

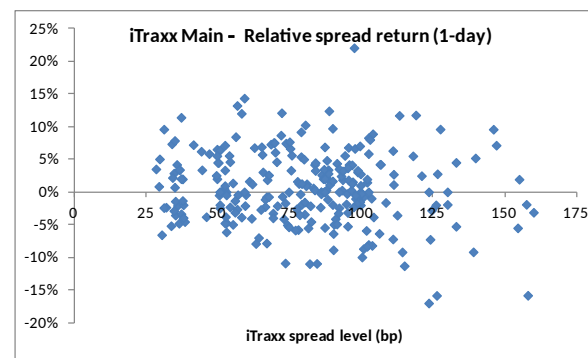
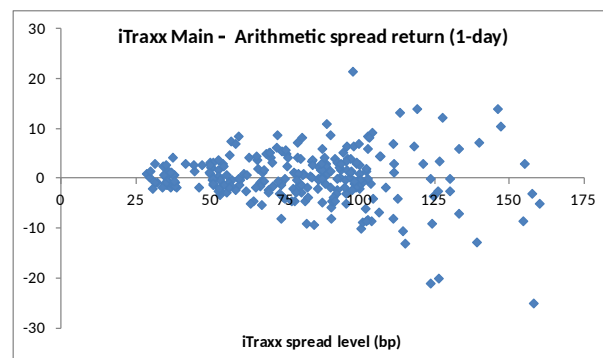
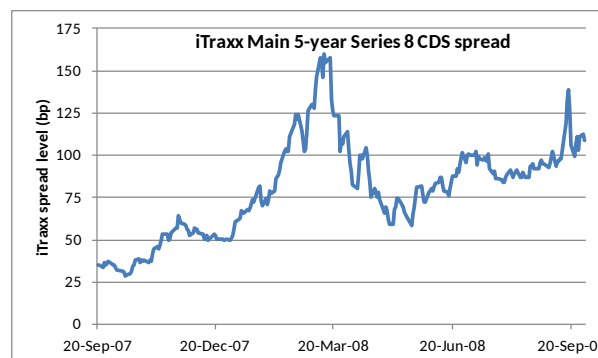
A common theme in using historical data to calibrate a simulation model is to remove the dependency on the then-pertaining market environment in past returns. This manifests itself in two ways: *which data* to use and *how* to use it.

■ Choice of relevant simulation dynamics

- The goal is to calibrate a model for future risk factor realizations based on past risk factor realizations
- Applying past risk factor realizations in today's environment can be done in several ways a priori

Classical example: The choice of return type affects the influence of the past market environment

- arithmetic difference
- relative return (or log-return)



Which type of return is more appropriate ?

- VaR requires the calculation of series of PnLs relative to today's market value.
- There are three distinct types of P&L calculations at JPMC:
 - Full revaluation (uses front-office pricing models; requires repricing of the entire portfolio)
 - Sensitivity based approach
- The sensitivity based approach uses spot Greeks in a Taylor expansion, which by nature is an approximation, but generally the terms included capture the material P&L.
- Typically, first order and some second order sensitivities are included.
- There are various other models in between Full reval and Sensi based models in market. These models are basically based on tradeoff between accuracy of Full reval and speed of sensitivity based models

Set-up:

- d portfolios with P&L represented by $L_1, \dots, L_d$ $L := L_1 + \dots + L_d$
- risk measure ρ

Often-seen phenomenon in practice: diversification effect, meaning that $\rho(L_1 + \dots + L_d) < \rho(L_1) + \dots + \rho(L_d)$

Capital allocation problem: How to define $K_1, \dots, K_d$ such that $K_1 + \dots + K_d = \rho(L)$

- proportional to standalone risk? $K_i = \frac{\rho(L_i)}{\sum_{j=1}^d \rho(L_j)} \times \rho(L)$
- proportional to marginal risk? $K_i = \frac{\rho(L) - \rho(L - L_i)}{\sum_{j=1}^d \{\rho(L) - \rho(L - L_j)\}} \times \rho(L)$

Fairness requirement: all portfolios (or arbitrary combinations of portfolios) should share in diversification benefit: $K_i \leq \rho(L_i)$

Theoretical results

If the risk measure satisfies $\rho(t \times L) = t \times \rho(L)$, $t > 0$, (positive homogeneity)
then it follows from Euler's homogeneous function theorem that $\rho(L) = \sum_{i=1}^d \frac{\partial \rho(L)}{\partial \lambda_i}$.

Proof: Define $L_\lambda = L(\lambda) := \sum_{i=1}^d \lambda_i L_i$, $\lambda \in \mathbb{R}^d$; Then Euler theorem says that $\rho(L_\lambda) = \sum_{i=1}^d \lambda_i \frac{\partial \rho(L_\lambda)}{\partial \lambda_i}$.

It follows that $\rho(L) = \rho(L_{(1, \dots, 1)}) = \sum_i \frac{\partial \rho(L)}{\partial \lambda_i}$.

Moreover, if ρ additionally is *coherent*, then the Euler principle is the only fair allocation scheme. Thus, allocation based on marginal contributions is sensible (and introduces the right incentives).

Challenges

1. Missing data or Frequency mismatch of historical data
 - Market data for recently introduced products
 - Illiquidity or non-daily remarking of curves or surfaces
 - Macroeconomic data (unemployment rate, home prices) arrive at a low frequency
2. Discrepancy between short-term behavior and long-term behavior of data series
3. Serial correlation in returns

Potential Solutions

Frequency mismatch of historical data

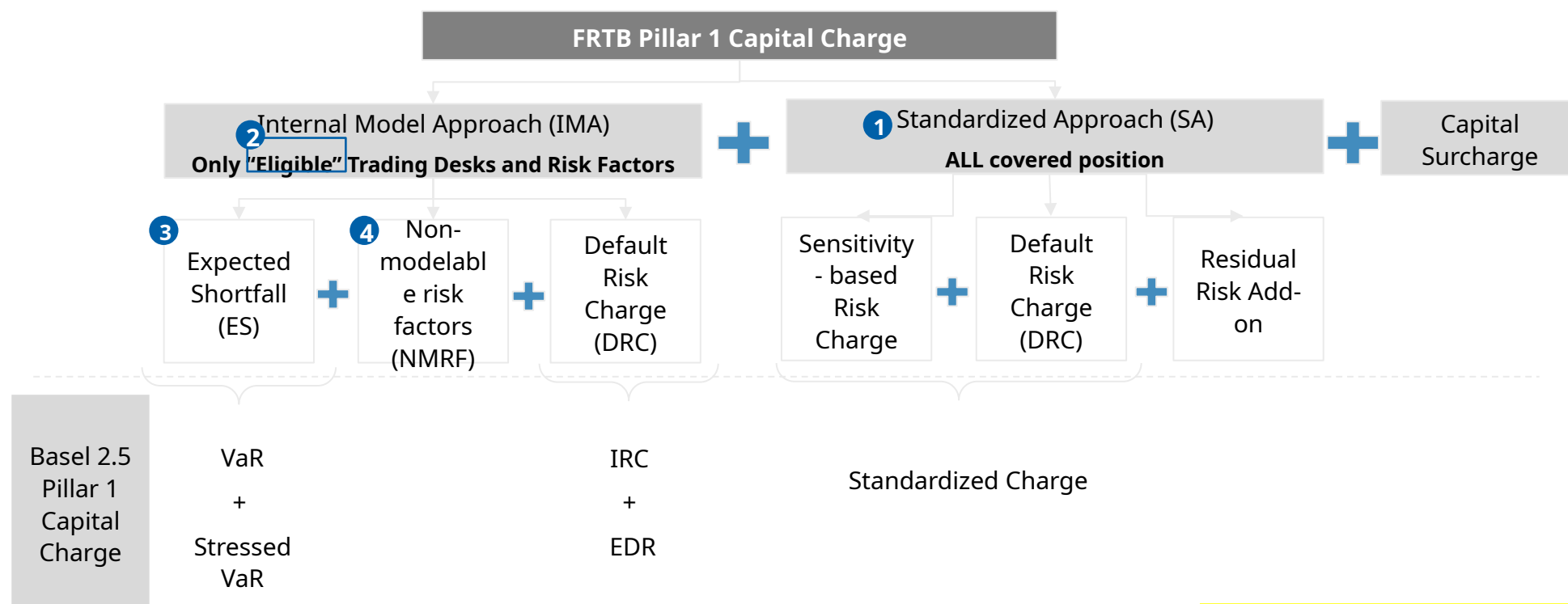
- Discard all zero-returns and treat the remaining returns as cumulative since the last return (“sticky method”)
- Treat missing data points as latent variables, and use the Expectation-Maximization (EM) algorithm to iteratively find the maximum likelihood estimate of the parameters of the assumed distribution
- Proxy the risk factor (May introduce unwanted basis)

Correlation:

- Decomposition of data series into high- and low-frequency components (wavelet analysis)

Fundamental Review of the Trading Book (FRTB) Overview

FRTB is the next generation market risk regulatory capital rule replacing the current Basel 2.5 framework. It introduces prescribed model performance tests to determine the Trading Desk eligibility for the internal model based approach with a defined standardized approach based charge applicable to all positions as a fallback. It overhauls Basel 2.5 capital components with more focus on tail risk, risk factor observability, prescribed liquidity horizon and provides requirements on calibration, aggregation and diversification recognition.



- Securitizations are not allowed to use IMA in FRTB
- 1-day VaR will still be required for backtesting and Risk Management VaR
 - CRM, RNIV and de minimis charge will no longer be required

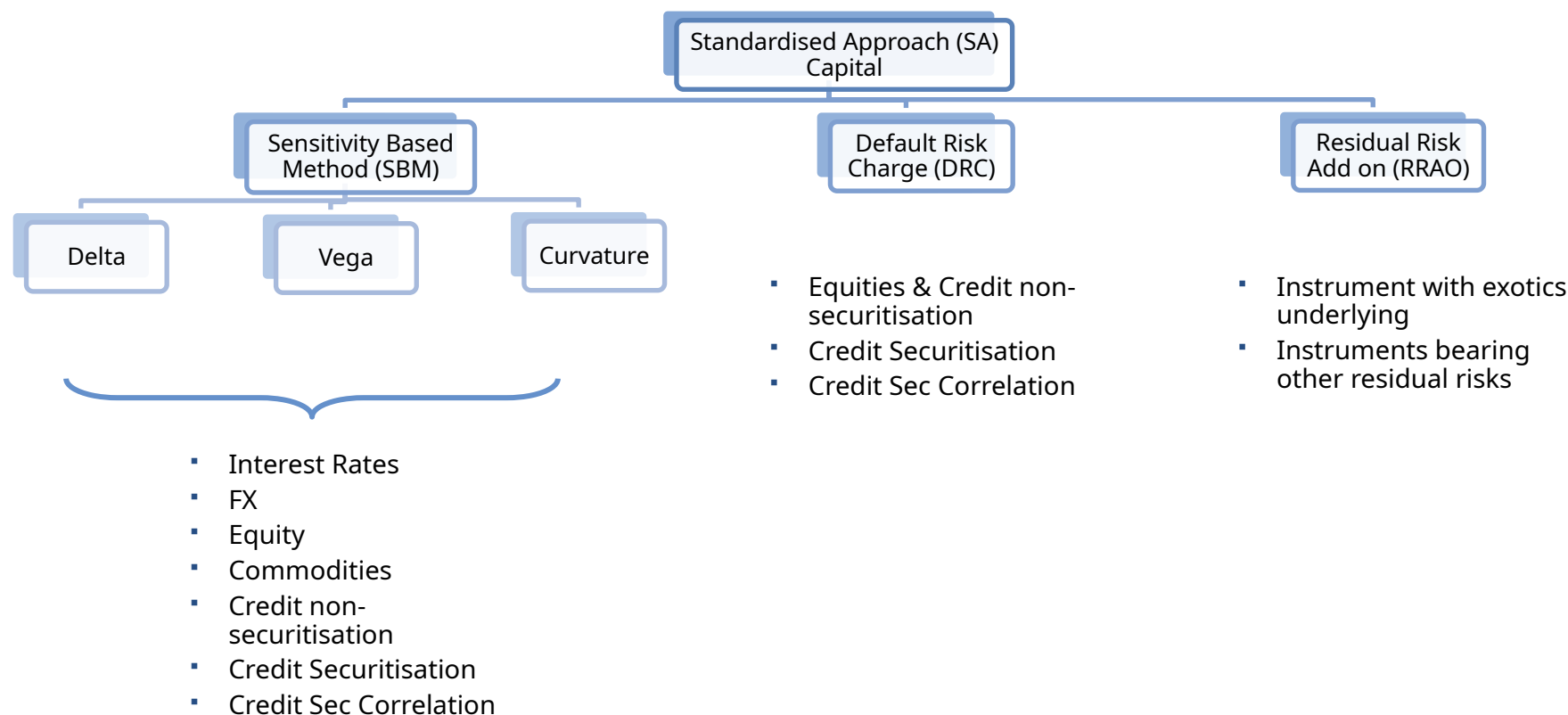
→ Refer to the following slides for additional details on key concepts

Standardized Approach Overview

Calculation of Capital charge under FRTB SA

Capital = Sensitivity Based Method + Default Risk Charge + Residual Risk Add On

- Spot based capital as of each month-end / quarter-end reporting (US Proposed Rule implies weekly)
- Prescribed risk weights / shocks that have been calibrated by the regulators to the losses seen in stress periods
- Sensitivity Based Method is max (Delta + Vega + Curvature) capital across three prescribed correlation scenarios (high, medium, low)
- Aggregation Mechanism for Sensitivity Based Methodology
 - Netting of exact matching risk factors
 - Aggregation with other risk factors using prescribed correlation parameters and via Variance-Covariance formula
 - Simple (undiversified) sum across asset classes



Standardized Approach: Delta and Vega Capital – Definition, Aggregation & Correlation

Delta & Vega Capital Charge

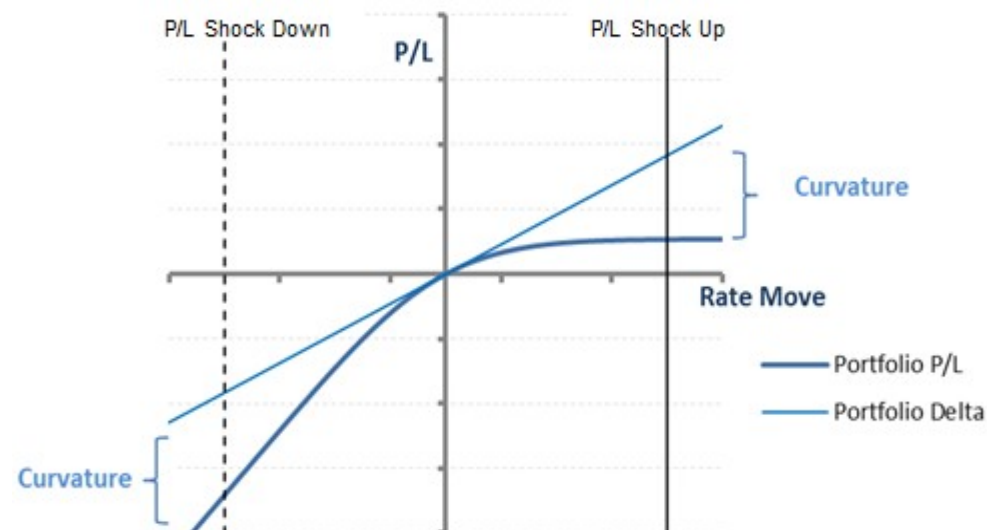
- Weighted Sensitivity calculation is based on prescribed risk weights and buckets which are applied to sensitivities
- Weighted sensitivities are then aggregated using prescribed intra bucket correlations to get the Capital at bucket level
- The last step to get to the overall Capital Charge using cross buckets correlation

Building Blocks	SA Delta Capital Example (GIRR)	SA Vega Capital Example																														
Sensitivities ×	$S_{i,r_t} = \frac{V_i(r_t + 1 \text{ bp}, cs_t) - V_i(r_t, cs_t)}{1bp}$	$S_k = Vega \times Implied \ Volatility$																														
Risk Weights =	Delta and Vega risk weights are prescribed by the rules*																															
Weighted Sensitivity ×	$WS_k = RW_k S_k$																															
Intra-bucket Correlations (correlation between risk factors within a bucket) =	<table><tr><th>TENOR</th><th>0.25</th><th>0.5</th><th>1</th><th>2</th></tr><tr><th>0.25</th><td>100.0%</td><td>97.0%</td><td>91.4%</td><td>81.1%</td></tr><tr><th>0.5</th><td>97.0%</td><td>100.0%</td><td>97.0%</td><td>91.4%</td></tr><tr><th>1</th><td>91.4%</td><td>97.0%</td><td>100.0%</td><td>97.0%</td></tr><tr><th>2</th><td>81.1%</td><td>91.4%</td><td>97.0%</td><td>100.0%</td></tr><tr><th>3</th><td>71.9%</td><td>86.1%</td><td>94.2%</td><td>98.4%</td></tr></table>	TENOR	0.25	0.5	1	2	0.25	100.0%	97.0%	91.4%	81.1%	0.5	97.0%	100.0%	97.0%	91.4%	1	91.4%	97.0%	100.0%	97.0%	2	81.1%	91.4%	97.0%	100.0%	3	71.9%	86.1%	94.2%	98.4%	GIRR : $\rho_{kl} = \min[\rho_{kl}^{(option \ mat)} * \rho_{kl}^{(underlying \ mat)}; 1]$ Other : $\rho_{kl} = \min[\rho_{kl}^{(DELTA)} * \rho_{kl}^{(option \ mat)}; 1]$
TENOR	0.25	0.5	1	2																												
0.25	100.0%	97.0%	91.4%	81.1%																												
0.5	97.0%	100.0%	97.0%	91.4%																												
1	91.4%	97.0%	100.0%	97.0%																												
2	81.1%	91.4%	97.0%	100.0%																												
3	71.9%	86.1%	94.2%	98.4%																												
Capital per bucket ×	$K_b = \sqrt{\max(0, \sum_k WS_k^2 + \sum_k \sum_{k \neq l} \rho_{kl} WS_k WS_l)}$																															
Inter-bucket Correlations (correlation across buckets within a risk class) =	$\gamma_{bc} = 0.5$																															
Capital Charge	Linear Risk = $\sqrt{\sum_b K_b^2 + \sum_b \sum_{c \neq b} \gamma_{bc} S_b S_c}$ 5																															

Standardized Approach: Curvature Capital – Definition, Aggregation & Correlation

Curvature

- Calculated only for products with **optionality**, and intended to capture **higher order risks**
 - Instruments with embedded options (callable bonds, correlation products, prepayment etc.) also require curvature capital
- It is the difference between ‘**Stress P&L based on prescribed risk weight**’ and ‘**Risk Weight x delta**’.
- Curvature Risk is based on **two stress scenarios** involving an upward shock and a downward shock to each regulatory risk factor
- No offset allowed for positive gamma** (or convexity) on delta losses; misaligns capital outcome vs risk management



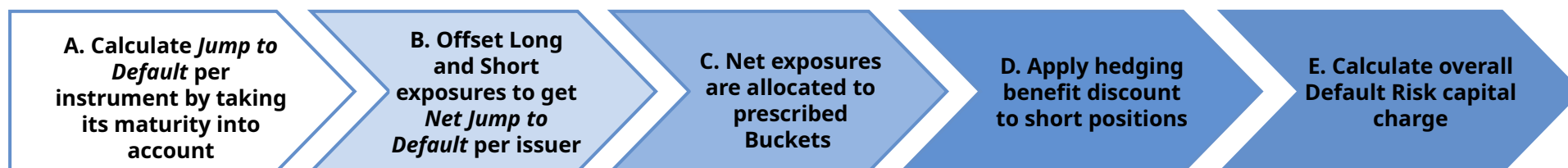
	SA Curvature Capital Building Block	Example Details
Stress P&L – Risk Weighted Delta	Curvature Risk Charge for each risk factor	$CVR_k^+ = -(\sum_i V_i (x_k^{RW(Curvature)^+}) - V_i(x_k) - RW_k^{Curvature} * S_{ik})$ $CVR_k^- = -(\sum_i V_i (x_k^{RW(Curvature)^-}) - V_i(x_k) + RW_k^{Curvature} * S_{ik})$
Aggregation Step 1	Curvature Risk exposure aggregation within each bucket*	$K_b \equiv \max(K_b^+, K_b^-)$
Aggregation Step 2	Curvature Risk exposure aggregation across buckets*	$Curvature\ capital = \sqrt{\max(0, \sum_b K_b^2 + \sum_b \sum_{c \neq b} \gamma_{b,c} S_b S_c \Psi(S_b, S_c))}$

*See Appendix 3 for K_b calculation and formula factor keys.

Standardized Approach: Default Charge

The **Default Risk Capital (DRC)** requirement is intended to capture **Jump-to-Default (JTD)**, a risk of a sudden default that may not be captured by credit spread shocks under the sensitivities-based method.

Example: Default Risk Charge for **equities and non-securitizations** is computed using the following steps:



- A. To calculate the risk weighed Jump To Default, risk weights are determined by **category buckets** and **credit ratings**. The **Jump To Default** exposures are scaled by maturity of the instrument (scaling = maturity / 1y; min = 0.25 and max = 1)
- B. Netting of long and shorts is based on **seniority** ie. only short instruments with lower (or same seniority) can net a long instrument
- C. Net Jump To Default positions are allocated to each asset class specific **buckets** determined by the rules
- D. To recognize a hedging relationship between net long and net short position within a bucket, a **hedge benefit** is computed

$$WtS = \frac{\sum net JTD_{Long}}{\sum net JTD_{long} + \sum |net JTD_{Short}|}$$

- E. DRC charge is **calculated separately for each bucket** (corporates, sovereigns, local governments / municipalities and defaulted) and then added together

$$DRC_b = \max[(\sum_{i \in Long} RW_i * net JTD_i) - WtS * (\sum_{i \in Short} RW_i |net JTD_i|); 0]$$

Standardized Approach: RRAO

Residual risk add-on (RRAO) is a **Notional Based charge** meant to capture risks that are not captured elsewhere in the Standardized Approach.

These include instruments that are 'non-linear, non-vanilla options replicable and non-cleared/exchange traded'

OR

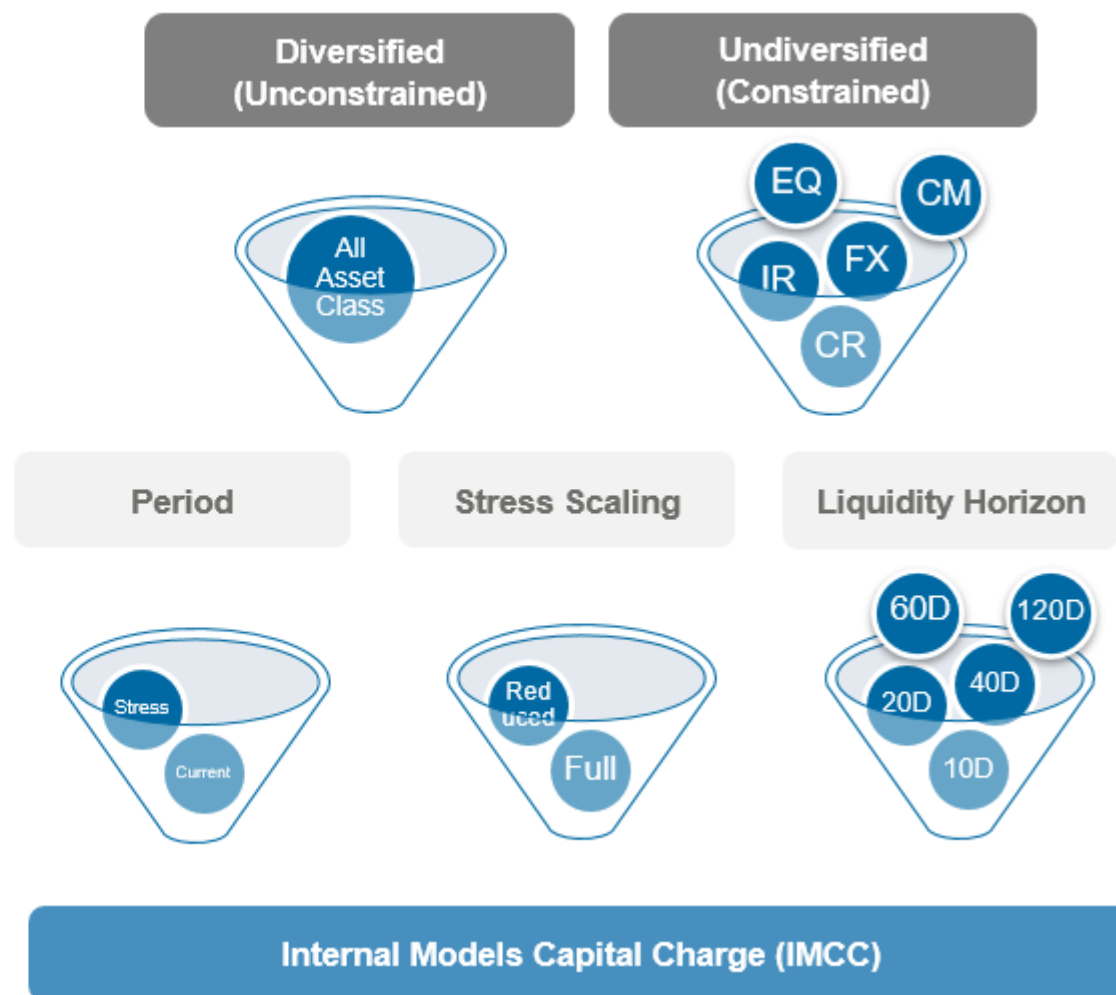
instruments with exotic underlying.

- Capital = Simple sum of "Notional x Risk Weight" for all instruments within scope of residual risk.
- The Risk Weight is defined as:
 - **1%** for **instruments with Exotic underlying**. For example, Longevity Risks, Weather Derivatives, Natural disasters
 - **0.1%** for instruments with **Other Residual risks**. Some examples are:
 - Path dependent (or non-vanilla replicable) options e.g., Bermudan
 - Correlation trading positions with three or more underlyings e.g., securitization tranches
 - Behavioural risks e.g., Mortgages (that have risk of pre-payments)
- Rates CMS spread options have been excluded from Residual Risk capital in US rules.

Internal Model Approach - Overview

The Internal Models Capital Charge (IMCC) is a weighted average of the cross-asset diversified Expected Shortfall and the sum of the standalone Expected Shortfall for each of the asset classes (Commodities, Credit, Rates, FX, Equities).

- Regulatory **Expected Shortfall** (ES) is a **multi-layered** calculation accounting for liquidity horizon and stress.
- The Expected Shortfall (ES) is to be calculated using **true 10D returns** at a **97.5% confidence level** (99% VaR $\approx$ 97.5% ES).
- **Stress Scaling** uses a “volatility scaling” approach for risk factors which do not have sufficiently long history of data for ES in stressed period.
- FRTB rule introduced **liquidity horizon scaling** to capture liquidity risk associated with different risk factors cross asset classes.
- The weight applied between Cross-Asset Diversified ES and sum of asst class ES is **50%**.



Internal Model Approach – Stress Scaling and IMCC Aggregation

Stress scaling calibration uses a “volatility scaling” approach for risk factors which do not have a sufficiently long history of market data observations going back to Stress period.

$$ES = ES_{R,S} \times \max\left(\frac{ES_{F,C}}{ES_{R,C}}, 1\right)$$

$ES_{R,S}$ is the bank's **expected shortfall** calibrated to the most severe **stress period** using proxies for risk factors where market data does not go back to a minimum period up to 2007.

The rule refers to the use of proxies as the **reduced set**.

$ES_{R,C}$ is the bank's **expected shortfall** for the **current period** for the **reduced set**.

$ES_{F,C}$ is the bank's **expected shortfall** for the **current period** for the **full set** of risk factors

The reduced set of risk factors must be able to explain a minimum of 75% of the variation of the full Expected Shortfall Model.

$$1 - \frac{\sum_t^{60} (ES_{F,C,t} - ES_{R,C,t})^2}{\sum_t^{60} (ES_{F,C,t} - \text{Mean}(ES_{F,C}))^2} \geq 75\%$$

Internal Model Capital Charge is a weighted average of the **cross-asset diversified ES** and the **sum of standalone measures** for each of the asset classes. GIRR, CSR, FX, Equity, Commodity. Currently, the diversification parameter ρ is set to **0.5**

$$IMCC = \rho(IMCC(C)) + (1 - \rho)\left(\sum_{i=1}^R IMCC(C_i)\right)$$

Diagram illustrating the IMCC formula components:

- Total Capital** points to $IMCC$.
- Fully Diversified Capital** points to $IMCC(C)$.
- Capital by Asset Class** points to $\sum_{i=1}^R IMCC(C_i)$.
- Diversification Parameter** points to ρ .

Internal Model Approach – Capital Calculation

Non-Modellable Risk Factors (NMRF) are risk factors that lack sufficient real price data for modelling purposes. The standards specify a separate stress scenario risk measure for each non-modellable risk factor.

Stressed expected shortfall (SES) capital for non-modellable risk factors requirements:

- Calibrated to 97.5% Expected Shortfall
- Measured at 20 liquidity horizon or greater

Idiosyncratic (ISES) credit spread or equity risk factors could be aggregated in non-modellable risk factor capital charge with 0 correlation subject to supervisory approval. Other risk factors are aggregated using a 60% correlation ρ .

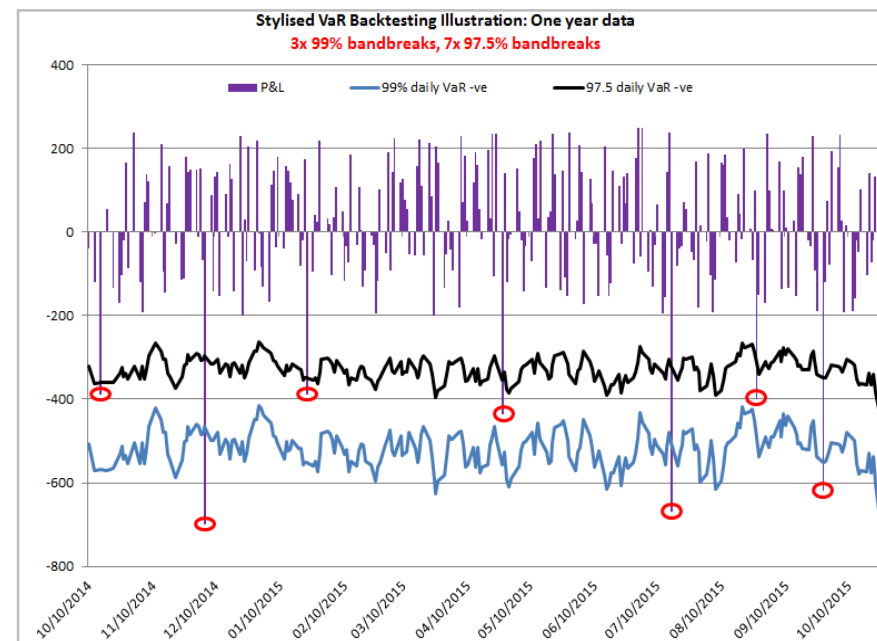
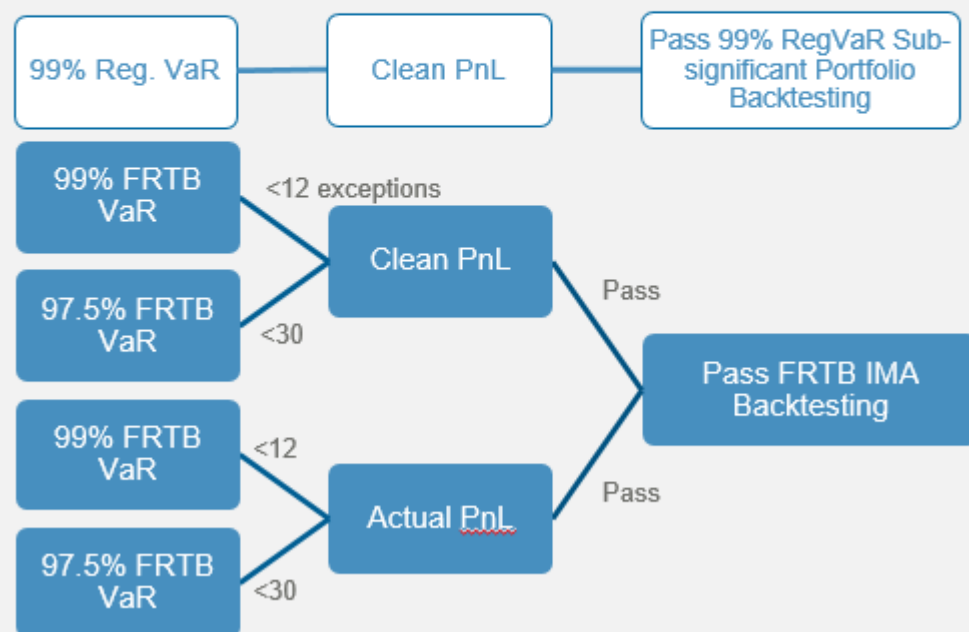
$$SES = \sqrt{\sum_{i=1}^I ISES_{NM,i}^2} + \sqrt{\sum_{j=1}^J ISES_{NM,j}^2} + \sqrt{\left(\rho * \sum_{k=1}^K SES_{NM,k}\right)^2 + (1 - \rho^2) * \sum_{k=1}^K SES_{NM,k}^2}$$

For the aggregated charge for IMA-approved Desks reported, capital from IMA (ex DRC) is the maximum of the spot value and daily average over the prior 60 days (with entity-wide backtesting multiplier m_c)

$$IMA_{G,A} = DRC_{SA} + \left(\max \left((IMCC_{t-1} + SES_{t-1}), \left((m_c \times IMCC_{average}) + SES_{average} \right) \right) \right)$$

Internal Model Approach - Backtesting

- VaR for each desk is back-tested against official P&L* for the **prior 12 months** at a desk level
 - 1 day VaR at 97.5% and 1 day VaR at 99% vs. clean (hypothetical) and actual P&L of the desk.
 - Exceptions for actual losses are counted separately from hypothetical losses.



Internal Model Approach – P&L Attribution

P&L Attribution Tests consist of two statistical tests performed quarterly using daily data for the prior 12 months to measure the alignment of the front office clean P&L and the equivalent P&L using risk models.

- Risk Theoretical P&L (RTPL): P&L calculated using risk models aligned with VaR and using front office aligned market data.
- Hypothetical P&L (HPL): The official clean P&L calculated using front office pricers (i.e. equivalent to current “clean P&L”)

Objective is to measure alignment to ensure sufficiency of risk factor coverage and accuracy of the pricers in the risk model.

Test 1: Spearman Rank Correlation

Measures ordered, pair-wise correlation between Risk Theoretical P&L and Hypothetical P&L
(note backtesting is only a distributional test)

Zone	Range
Green	$SR \geq 0.80$
Amber	$0.70 \leq SR < 0.80$
Red	$SR < 0.70$

Test 2: Kolmogorov-Smirnov Statistic

Tests the null hypothesis that Theoretical P&L and Hypothetical P&L are both samples of the same (unspecified) underlying distribution

Zone	Range
Green	$KS \leq 0.09$
Amber	$0.09 < KS \leq 0.12$
Red	$KS > 0.12$

A desk is in the red zone if either test is **not satisfied**
Overall zone determined by the **lowest zone** of the two tests.

Desk Status	Internal Models Approach Eligibility Criteria
Green	IMA eligible
Amber	IMA eligible, but with capital surcharge ('PLA add-on')
Red	Capitalized under SA with other IMA ineligible desks